

Financial Chatbot Prototype Using Python

This project involved creating a simple AI-powered financial chatbot that analyzes financial data and responds to predefined queries. The process was broken down into two main tasks:

1. Data Analysis:

- I started by extracting financial data from 10-K reports, specifically focusing on key metrics such as total revenue and net income.
- Using Python and libraries like pandas, I imported the data from a CSV file and converted it into a DataFrame for analysis.
- Calculations like revenue growth and net income growth were performed, providing valuable financial insights.

2. Chatbot Development:

- A basic chatbot was developed using Python, where predefined queries were mapped to specific responses based on the financial data analysis.
- The chatbot logic used simple if-else statements to respond to queries such as "What is the total revenue?" and "How has net income changed over the last year?".
- Flask was incorporated to potentially create a web-based interactive version of the chatbot, making the tool more accessible.

Implications:

This project demonstrates how AI-powered tools can simplify complex financial data analysis and make insights more accessible. It highlights the potential for integrating AI and financial analysis, which can be particularly valuable in roles that involve data-driven decision-making, like those in banking and finance.